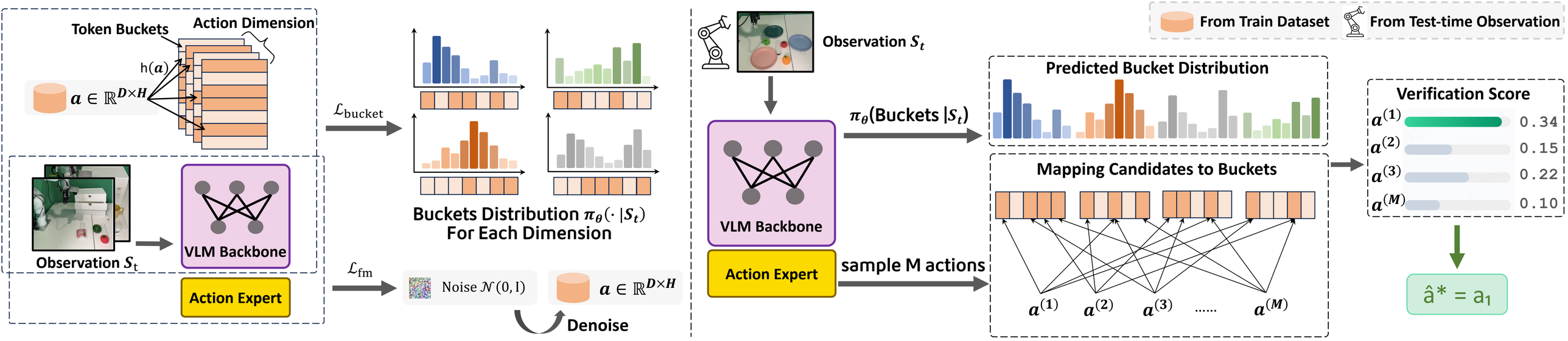

Token Buckets
Action Dimension
h(a)
$a \in \mathbb{R}^{D\times H}$
Observation S_t
VLM Backbone
Action Expert
$\mathcal{L}_{\text{bucket}}$
Buckets Distribution $\pi_\theta(\cdot\,|S_t)$
For Each Dimension
$\mathcal{L}_{\text{fm}}$
Noise $\mathcal{N}(0, \mathrm{I})$
$a \in \mathbb{R}^{D\times H}$
Denoise
(a) Training Stage
From Train Dataset
From Test-time Observation
Observation S_t
VLM Backbone
Action Expert
π_θ(Buckets $|S_t$)
sample M actions
Predicted Bucket Distribution
Mapping Candidates to Buckets
$a^{(1)}$ $a^{(2)}$ $a^{(3)}$ ..... $a^{(M)}$
Verification Score
$a^{(1)}$ 0.34
$a^{(2)}$ 0.15
$a^{(3)}$ 0.22
$a^{(M)}$ 0.10
â* = a1
(b) Inference Stage